Git and GitHub Complete Study Notes

1. Version Control

Version Control System (VCS) records every change made to files over time. It allows developers to track history, collaborate safely, compare versions, and restore older versions when needed.

2. Git

Git is a distributed version control system installed on your computer. It tracks changes locally, works offline, stores commits, supports branching, merging, and collaboration.

3. GitHub

GitHub is a cloud platform that hosts Git repositories. It is used for backup, collaboration, pull requests, code reviews, issue tracking, and sharing projects.

4. Git vs GitHub

Git is the software that manages version history locally. GitHub is the online service where Git repositories are stored and shared.

5. Git Workflow

Working Directory → Staging Area → Local Repository → Remote Repository (GitHub).

6. Basic Commands

git init - Initialize repository.

git status - Show repository status.

git add / git add . - Stage changes.

git commit -m "message" - Save snapshot.

git log / git log --oneline - View commit history.

git diff - Show changes before commit.

7. Branching

A branch is an independent line of development. Main branch stores stable code. Commands: git branch, git switch, git switch -c, git merge, git branch -d. Branches allow multiple developers to work independently. Merge conflicts occur when two branches modify the same code and must be resolved manually.

8. GitHub Remote Commands

git clone - Download repository.

git remote add origin - Connect local project.

git remote -v - Show remotes.

git push origin main - Upload commits.

git push -u origin main - First push and set upstream.

git pull origin main - Download and merge.

git fetch - Download changes without merging.

9. Pull vs Fetch

git fetch downloads updates only. git pull downloads and automatically merges them into the current branch.

10. Local vs Remote Repository

Local repository exists on your computer and works offline. Remote repository is stored on GitHub and is used for collaboration and backup.

11. Repository

A repository contains all project files and Git history.

12. Team Workflow

Developer creates feature branch → develops feature → commits → pushes to GitHub → creates Pull Request → code review → merge into main.

13. Best Practices

Write meaningful commit messages.

Create feature branches.

Pull before pushing.

Use .gitignore.

Commit small logical changes.

Keep main branch stable.

14. Common Interview Questions

• What is Git? Distributed Version Control System.

• What is GitHub? Cloud platform for Git repositories.

• Difference between Git and GitHub.

• What is a branch?

• Difference between fetch and pull.

• What is a commit?

• What is a repository?

15. Quick Revision

Git = Version control

GitHub = Cloud hosting

Repository = Project

Commit = Snapshot

Branch = Separate workspace

Merge = Combine branches

Push = Upload

Pull = Download & merge

Fetch = Download only

Clone = Copy repository